Open source · free-tier cloud

FerroFlux

Think Steel. Think Smart.

Intelligent Steel Supply-Chain Analytics Platform

Digilians · Nano Degree (4 Months) · Track: AI & Data Science — Professional Big Data Engineer

1 Project Identity

Project	FerroFlux Platform
Supervisor	Dr. Rania ElGohary
Track Head	Dr. Mohamed El Badawi
Instructor	Dr. Amera Youssef

2 Team

- Nourhan Saber Shalaby
- Hager Elkahlawy
- Haidy Nageh
- Maram Mostafa
- Zeinab Elgabalawy

3 Problem & Audience

86,044	3	12
rows of data	factories	suppliers

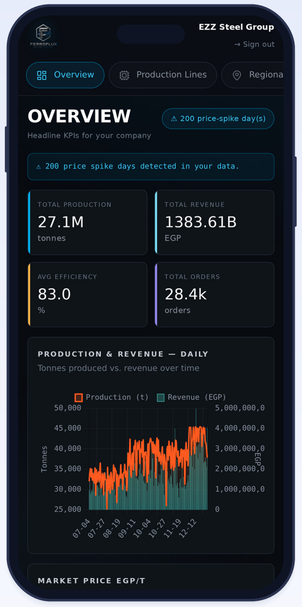
Simulated EZZ Steel Group dataset

Fragmented spreadsheets and manual price tracking

Users: factory managers, admins, decision makers

4 Solution & Core Idea

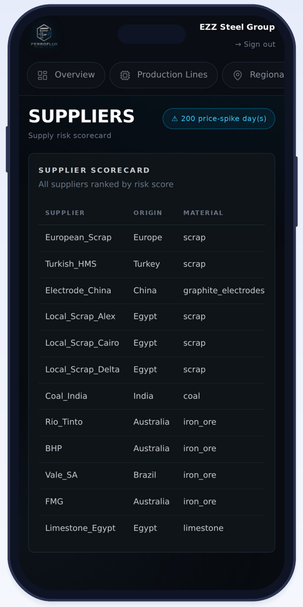
One platform that streams, cleans and predicts — and automatically alerts each factory to price spikes and low efficiency.



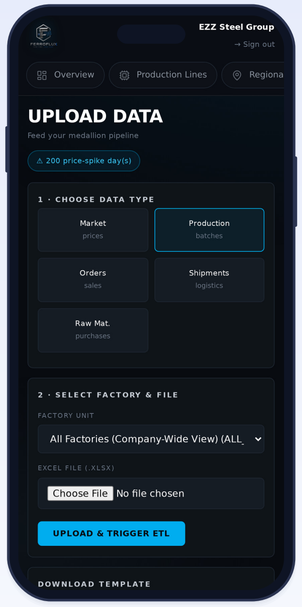
1 Tenant overview



2 Spike & demand analytics



3 Supplier risk scorecard



4 Excel upload to ETL

SYSTEM ARCHITECTURE

Stream · Predict · Alert

1 SOURCES	Factory telemetry	Market price feed	Excel uploads			
2 STREAM & ETL	Apache Kafka	Spark: Bronze › Silver › Gold	Airflow	Anomaly detection		
3 STORE & ML	PostgreSQL	Neon	Price forecast	Demand forecast	Supplier risk	ALERT PATH
4 SERVE & ALERT	FastAPI	Streamlit	n8n	Docker	Hugging Face	Email + portal alerts

5 Pipeline

Ingest (Kafka) → Clean (Silver) → Detect & Alert → Aggregate (Gold x6) → Predict (3 models)

6 Key Features

- **Tenant isolation** — company_id on every read
- **Excel to auto-ETL** — upload triggers Airflow
- **Self-service signup** — register, reset by email
- **Spike alerts** — email, inbox, n8n
- **3 ML forecasts** — price, demand, risk
- **Live streaming** — Kafka to Spark

7 Why FerroFlux

	FerroFlux	SAP IBP	Tableau + Excel	Generic BI
Real-time streaming	✓	✗	✗	◑
Built-in ML forecasts	✓	✓	✗	◑
Automated alerts	✓	✓	✗	◑
Steel-industry focus	✓	✗	✗	✗
Self-service signup	✓	✗	✗	◑
Deployment cost	Free tier	Very high	Medium	High

8 Impact & KPIs

R^2 0.9578 — Price forecast R^2	MAPE 1.61% — Price forecast error
94.78% — Supplier-risk accuracy	R^2 0.80 — Demand forecast R^2

PROJECT LINK & QR CODE

Live portal · scan or open

huggingface.co/spaces/nourhan44/ferroflux-portal

Digilians · Graduation Project

Digital Pioneers Initiative · 2026